Md. Shahriar Ahmed

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Academic Credentials

University of Northern Iowa Cedar Falls, IA 50614, USA
Masters in Geography August, 2025 - Present
CGPA : 4.00 / 4.00 (Current)
"A Spatial Framework for Social, Cultural, and Physical Asset Valuation and Risk Assessment under Coastal Hazards in Alaska"

Rajshahi University of Engineering & Technology Rajshahi, Bangladesh
Bachelor in Urban & Regional Planning January, 2018 - October, 2023
CGPA : 3.59 / 4.00 (10th)
"The Economic Impact of Tropical Storms on the Sundarbans Mangrove Ecosystem Services and Local Communities"

Professional Experience

University of Northern Iowa, Department of Geography Cedar Falls, IA 50614, USA
Graduate Research Assistant at ARCTICenter August, 2025-Present

- NNA Collaboratory: ACTION - Alaska Coastal Cooperative for Co-producing Transformative Ideas and Opportunities in the North (Monitoring shoreline change in Nelson Lagoon using remote sensing and GIS, Additional research on fisheries governance and ecotourism suitability in northern coastal communities.)

Center for Environmental and Geographic Information Services (CEGIS) Dhaka, Bangladesh
Research Consultant at Remote Sensing Division November, 2023 - May,2025

- Advanced GIS Analysis on Remote Sensing Data (Spatial Analysis, Developing Geospatial Database analysis and Map preparation,Digital Database Management for Data Accumulation,Compilation of Land Use Zoning and Cropping Pattern).
- Chittagong City Water Logging Project (Develop and Renovate the Drainage Network by Connecting the Drains and Canals).
- Generation of the Land Cover and National Capital Map and Developing Integrated Collaborative Forest Management Plan.

Rajdhani Unnayan Kartipakkha (RAJUK) Dhaka, Bangladesh
Intern Planner March, 2023-April, 2023

- Preparation of Concept Plan for Gabtoli TOD Area: Northern part,Bangladesh.

Research Interests

My research focuses on addressing environmental challenges such as climate change, coastal hazards, land-use change, and natural disasters. I use GIS and remote sensing to analyze spatial patterns and support sustainable, resilient, and efficient systems that improve human well-being.

Scholarships and Awards

- **Bangladesh Government Scholarship** (*General)*: Higher Secondary Certificate,2017
- **Bangladesh Government Scholarship** (*General*): Secondary School Certificate,2015

Research Experience

Publication (Book Chapter)

- Ahmed,M.S., Rahman,M.A., Das,A.(2024). The Economic Impact of Tropical Storms on the Sundarban's Mangrove Ecosystem Services and Local Communities. *Environmental Economics and Policy for a Changing* . (in review)
- Ahmed,M.S., Rahman,M.A., Das,A.(2024). Balancing Conservation and Use: Managing Common Pool Resources in the Sundarbans Mangrove Forest. *Environmental Economics and Policy for a Changing* . (in review)

Conference Activities

Poster presentation

- Assessment of the impact of disasters on agricultural production and its influence on overall development. West Lakes Regional Division Meeting 2025.
- An analysis on the morbidity of slum dwellers: A case study of Rajshahi City. *I*nternational Conference on Chemical Engineering (ICCHE) 2020. (Paper Id: 036)

Paper presentation

- Assessing Spatiotemporal Trends of Shoreline Change in Alaska: Measuring the Impacts of Sea Level Rise in the Arctic. 2026 AAG Annual Meeting. (Presenting Author).
- Spatiotemporal Disparities in Heatwave Exposure Across Alaska. 2026 AAG Annual Meeting. (Presenting Author).
- Alaska's Coastal Communities and Eco-tourism: The Challenges and Opportunities for Sustainable Economy. 2026 AAG Annual Meeting. (Co-Author).
- Salmon Fisheries Governance: Understanding Multi-Level Structures and Horizontal Linkages that Support or Impede Indigenous Community Resilience in the North Pacific. 2026 AAG Annual Meeting. (Presenter).

Panelist

- Norbert Psuty Student Paper Competition: Coastal and Marine Specialty Group. American Association of Geographers Annual Meeting, San Francisco, California, March 20, 2026.

Academic Research

- Assessment of Economic Changes through Short run and Long run Regional Econometric Analysis: A case Study on Bagerhat District in Bangladesh.
- An Analysis of Economic Loss Due to Traffic Congestion in the Rail Crossings: A case study in Rajshahi City Corporation.
- A Participatory Appraisal for Community development through strengthening human rights and employment generation in Shiroil Colony, Rajshahi.

Projects

- Alaska Coastal Communities & Birds Web Map. Github
- Alaska Peninsula Communities: Population and Climate Trends Visualization. Github
- Wildfire evacuation behavior in an Alaska community using an agent-based modeling (ABM) approach in NetLogo. Github
- Analysis of CO Concentration in Alaska Using Sentinel-5P Data in GEE. GEE

Technical Skills

- Geospatial Tools : ArcGIS Pro, QGIS, NetLogo, Google Earth Engine, ENVI, Erdas Imagine, Web Mapping
- Analysis Software : SPSS, R
- Programming Language : Python,Machine Learning, Java Script, CSS, SQL
- Design and Modelling : Adobe Illustrator, AutoCAD, SketchUp, Lumion.

Affiliations

Associate Member, Bangladesh Institute of Planners (BIP). Membership ID: BIP2334

Standardized Test Scores

International English Language Testing System (IELTS) -14th September,2024

Overall Score	Listening	Reading	Writing	Speaking
7.0	8	7.5	6	7

Extra-curricular Activities

Advisor Rajshahi, Bangladesh
RUET Cricket Club August 2023 -Present
RUET Cricket Team became 4th in the Inter University Cricket Championship,2024.

Captain Rajshahi, Bangladesh
RUET Cricket Club October 2022 -July 2023
Lead the RUET team in the Inter University Cricket Championship,2023.

Organizing Secretary Rajshahi, Bangladesh
Pabna District Association,RUET October 2022 -September 2023

Captain Rajshahi, Bangladesh
URP Football Team,RUET October 2022 -September 2023
Lead the URP Football Team in Inter Department Tournament,2023.

Captain Rajshahi, Bangladesh
URP Cricket Team,RUET October 2022 -September 2023
Lead the Cricket Team in Inter Department Cricket Tournament,2023.

Linguistic Proficiency

1. English (Fluent working proficiency)
2. Bengali (Native)

Courses and Workshops

- Spatial Data Science_ The New Frontier in Analytics by ESRI.
- Digital Image Processing with OpenCV in Python by Geo University.
- Remote Sensing and Satellite Image Processing with the EOS Platform by Geo University.
- Coordinate Reference Systems, Projections, and Spatial Data Representation by Geo University.
- Introduction to Remote Sensing by Geo University.